Inha University in Tashkent
School of Computer Science and Information Engineering
Course Name: Object Oriented Programming 2
Semester and Year: Spring Freshman, 2026
Laboratory Assignment 6: Object-Oriented Programming in C++

Note: 1. Submission only via E-CLASS is acceptable (If you have any issue with E-CLASS, Please consult IT Dept.)

2. Comments should be included in your source code.

3. Do not plagiarize or use AI tools to complete the assignment. If a student is caught doing so, he/she will receive ZERO for the assignment.

4. Upload your completed laboratory exercise as a zip file including the appropriate files **(.cpp, and .h),** to the eClass system before the deadline. The zip file must be named with your **ID number.**

Exercise 6: University Management System

Objective: The university management wishes to extend its database to include employee and student information. Additionally, understand the concept of polymorphism.

Tasks:

1. **Create Base class:**
   - Define a class Person.
   - Declare the following protected members: Name(string), Age(int), Gender (char).
   - Define parameterized constructor.
   - Define a setPerson() member function that takes name, age and gender as parameters and validate that age is positive value and gender is either 'M' or 'F'.
   - Define a virtual show() member function that prints the name, age and gender.

- Declare virtual destructor.

2. **Create two derived classes:** Student and Teacher, both inheriting from the Person class(public inheritance).

   - Declare the following private members for Student class: IDnumber (int), subject(string) and grade(int).
   - Define parameterized constructor. Constructor of Student must call constructor of Person.
   - Define a setStudent() member function that takes IDnumber, subject and grade as parameters. Validate that grade must be between 0 and 100 and IDnumber positive values.
   - Define a show() member function that prints the IDnumber, subject and grade.
   - Declare the following private members for Teacher class: courseInfo(string).
   - Define parameterized constructor. Constructor of Teacher must call constructor of Person.
   - Define a setTeacher() member function that takes courseInfo as parameter.
   - Define a show() member function that prints the courseInfo.

In main():

- Create an array of base-class pointers. Let each pointer point to a created object, and then display the full details of each object.

Output:

```
Invalid gender for Ali. Gender set to U.
Invalid student ID for Ali. ID set to 0.
Invalid grade for Ali. Grade set to 0.
Invalid gender for Bekzod. Gender set to U.
----- Student Details -----
Name   : Ali
Age    : 20
Gender : U
ID No. : 0
Subject: Mathematics
Grade  : 0

----- Teacher Details -----
Name   : Salima
Age    : 40
Gender : F
Course : Object Oriented Programming

----- Student Details -----
Name   : Malika
Age    : 19
Gender : f
ID No. : 102
Subject: Physics
Grade  : 95

----- Teacher Details -----
Name   : Bekzod
Age    : 35
Gender : U
Course : Data Structures
```